

Finite Difference Derivatives

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1 Abstract

We explore the implementation of finite difference derivatives in Newton's Method for root finding in this report. Finite difference methods for approximating the derivative can be generalized computationally allowing us to use a general R function for any input function. There are several different finite difference methods including the forward difference, $f'(x) \approx \frac{f(x+\delta)-f(x)}{\delta}$, and the three point centered difference, $f'(x) \approx \frac{f(x+\delta)-f(x-\delta)}{2\delta}$. Newton's Method is an iterative root finding method that calculates the tangent line at a point x_i and sets the equation to zero to find x_{i+1} . A starting value x_0 is required. As this method includes finding the tangent line at a point, it follows that the derivative is required. In general, the point $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$. The convergence of Newton's Method is usually quadratic which is faster than the bisection method, although in certain cases the convergence is linear such as at multiple roots.

2 Newton's Method

2.1 Derivative Function

This is the derivative function using the forward difference method outlined in the abstract. It takes a function f and a step, δ , and returns a function that is the finite difference approximation of the derivative of the inputted function. Below is the R code for this function with a default δ of 0.001.

```
D <- function(f, delta=0.001) {  
  derivative <- function(x) {  
    (f(x + delta) - f(x)) / delta  
  }  
  return(derivative)  
}
```

2.2 Derivative Function Graphs

To test the function D , we used the input function $f(x) = xe^{-x^2}$. Below are the plots of $f(x)$ and $f'(x)$.

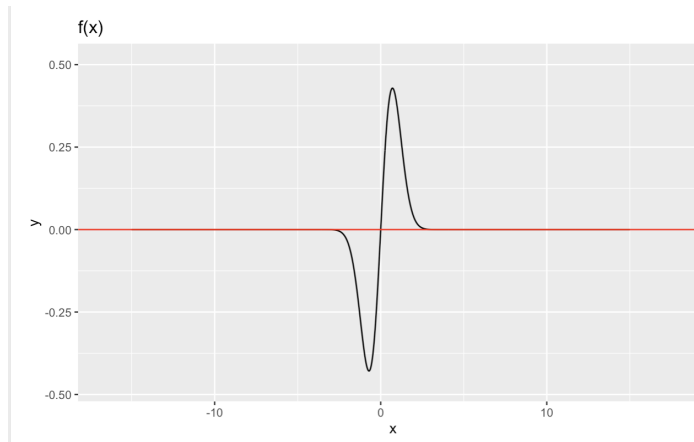


Figure 1: This is the graph of our initial function, $f(x) = xe^{-x^2}$.

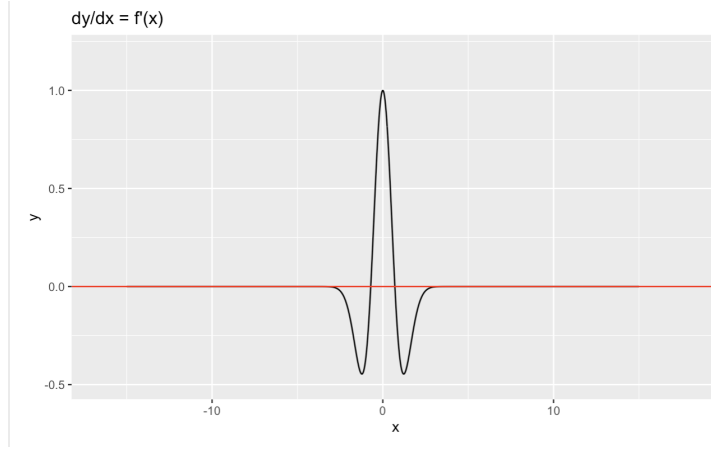


Figure 2: This graph here is the derivative graph of that same function. We used our R function to create a data frame with a huge range of deltas and their corresponding derivatives, and that data frame is plotted here.

2.3 Newton's Method

Newton's Method is an iterative root-finding procedure that utilizes the function's derivative at various x values until it reach a maximum amount of iterations or when the difference between the current and previous x values are within a certain threshold. For our implementation of Newton's Method, we chose to find a root after it reaches a certain distance threshold between the previous x values found. The method starts with an initial x -value and it finds the line tangent to that point on the original function. Then it finds when that line crosses the x -axis and that x -value at that point is our root or the next x -value to be used in the iteration, where we follow the same steps.

The slope of the line is found by the derivative of the function at the given x value. We can derive the equation to find the succeeding x value to get to our root with the point-slope formula with the slope of the line described previously, the x value as our initial guess, and the y value as the value of the function when evaluating for x . We would then want to solve for the general x where the general $y = 0$ is shown below.

$$(y - y_i) = m(x - x_i)$$

$$y - f(x_0) = f'(x_0)(x - x_0)$$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

$$0 = f(x_0) + f'(x_0)(x - x_0)$$

$$0 = f(x_0) + f'(x_0)x - f'(x_0)x_0$$

$$f'(x_0)x = f'(x_0)x_0 - f(x_0)$$

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

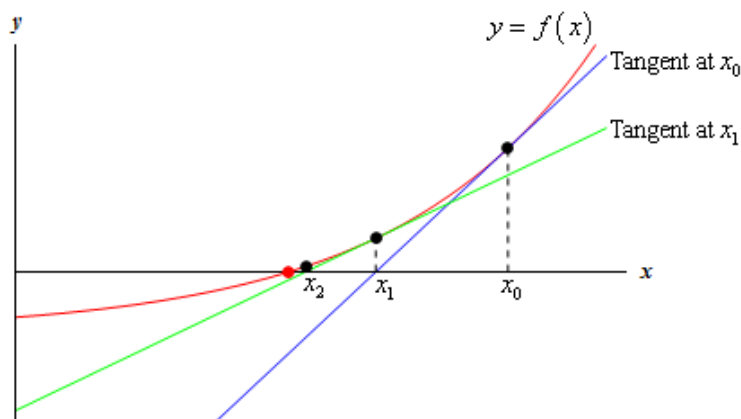


Figure 3: This image is a visual representation of Newton's Method for root finding where the red line is the original function, the x_0 is the initial guess of the root, and the blue and green line, respectively, are the following steps described above about finding the tangent line of the original function at different x -values.

2.4 Implementation

Here is our implementation of Newton's Method in R code. The function takes in a function, an initial starting guess, and is initiated with a default tolerance of 0.5×10^{-6} . It uses the derivative function called above, and then an initial evaluation with our imputed initial guess. It then leads into a while loop that continues to run until the difference in the current x value and the previous x value is less than our tolerance.

```
Newton <- function(f,x,tol=0.5*10^-6) {
  derivative <- D(f)
  x1 <- x - ((f(x))/(derivative(x)))
  while(abs(x1-x)>=tol){
    x <- x1
    x1 <- x - ((f(x))/(derivative(x)))
  }
  return(x1)
```

2.5 Breaking Newton's Method

There are some limitations to Newton's Method. For example, you can select initial x -values for certain functions that are so far away, that Newton's Method won't converge to the root. There is also a situation where Newton's Method "cycles" between points and never converges to zero as well. This is shown with the function of $f(x) = 4x^4 - 6x^2 - \frac{11}{4}$ and an initial guess of $x_0 = \frac{1}{2}$. When we use these parameters in our Newton's Method function, we get a never-ending loop because our answer never gets within the given tolerance.

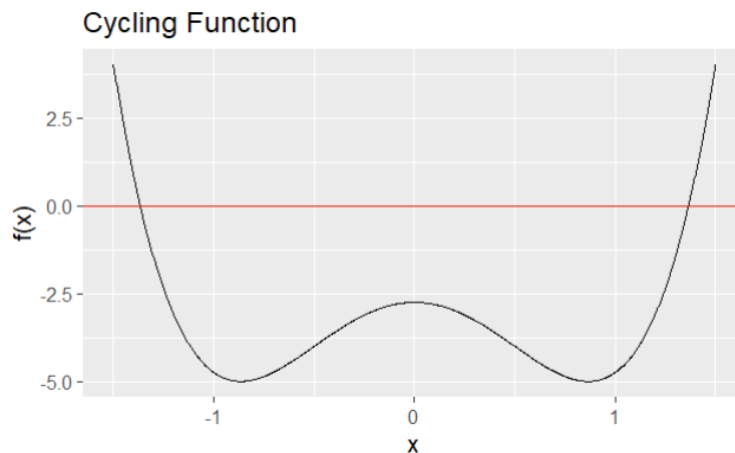


Figure 4: This is the graphical representation of the function $f(x) = 4x^4 - 6x^2 - \frac{11}{4}$ which would cause a cycling error with our Newton's Method function if the initial guess was $x_0 = \frac{1}{2}$.

3 Finite Derivatives

3.1 Difference in Deltas

The finite difference derivative approximation function that we implemented requires a specified step, delta. We cannot set delta to 0 because there would be a computational error that arises from dividing by 0. Here, we explore how the difference in delta impacts the accuracy of the approximation.

The plot below shows the approximated derivative of $f(x) = e^x$ using the forward difference function versus the the range of deltas used on the natural logarithm scale.

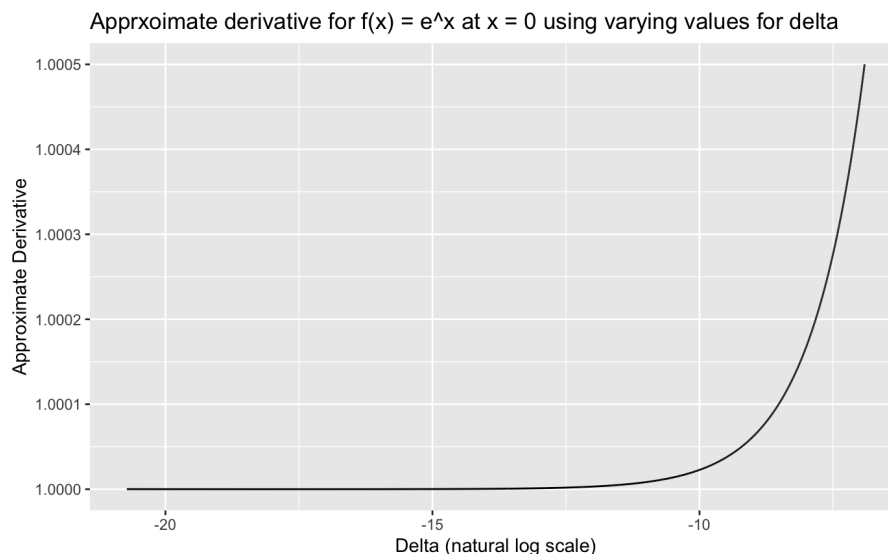


Figure 5: This is the plot of the function $f(x) = e^x$ at $x = 0$. This plots the approximated derivatives using the forward difference function versus the range deltas on the natural logarithmic scale. We see a steep line initially, indicating major change at each step, before a big flattening out just past $\text{delta} = e^{-10} = 0.00004539992$, as the plot approaches on exact limit of 1. It makes sense that this is the exact limit of the derivative for increasingly small values of delta because the derivative is defined as the limit as delta approaches 0 of the forward difference equation used in the derivative function earlier. The plot approaches 1 as the deltas on the logarithmic scale become increasingly negative which makes sense as the exact derivative of $f(x) = e^x$ at $x = 0$ is 1.

3.2 Difference in Approximation Methods

The finite difference approximation that is used has an impact on the errors of the solutions. The three point centered difference has errors that are magnitudes of order lower than the forward difference method. To show this, consider the function $f(x) = xe^{-x^2}$ that we used earlier. This function is nice because we know the derivative to be $f'(x) = e^{-x^2} - 2x^2e^{-x^2}$. At first glance, there does not appear to be any difference between the forward difference, the three point centered difference, or the exact solution based on the plots of the graphs $x \in (-5, 5)$. The plots for these three functions overlap each other and are present when the others are omitted.

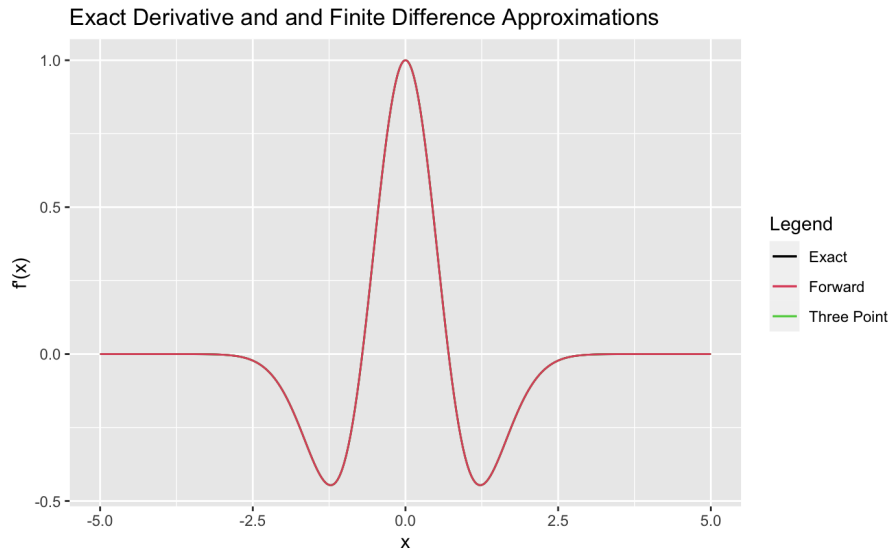


Figure 6: However, a closer examination of the errors of the forward difference method and the three point centered difference method reveals that the latter has errors magnitudes lower than the former. The plot below shows the difference between the forward difference and the exact solution, the difference between the three point centered difference and the exact solution, and the difference between the forward difference and three point centered difference all for $x \in (-5, 5)$. The error of the three point centered difference approximation remains constant and roughly zero in the domain. However, the error of the forward difference approximation spikes at roots of $f'(x)$ which are located at $x \approx 0.71$ and $x \approx -0.71$. While the error of the three point centered difference is not zero, it is orders of magnitude smaller than the error for the forward difference method which explains why it appears to be zero in the plot below. The points at which $f'(x) = 0$ are critical points which correspond to local maxima or minima on $f(x)$. It makes sense that the error for the forward difference method spikes at the extrema of $f(x)$ because the slope changes from positive to negative or negative to positive at these points and the forward difference method thus overestimates or underestimates the derivative at these points. This effect is not as noticeable in the three-point centered difference method because it takes a wider range of points to calculate the difference.

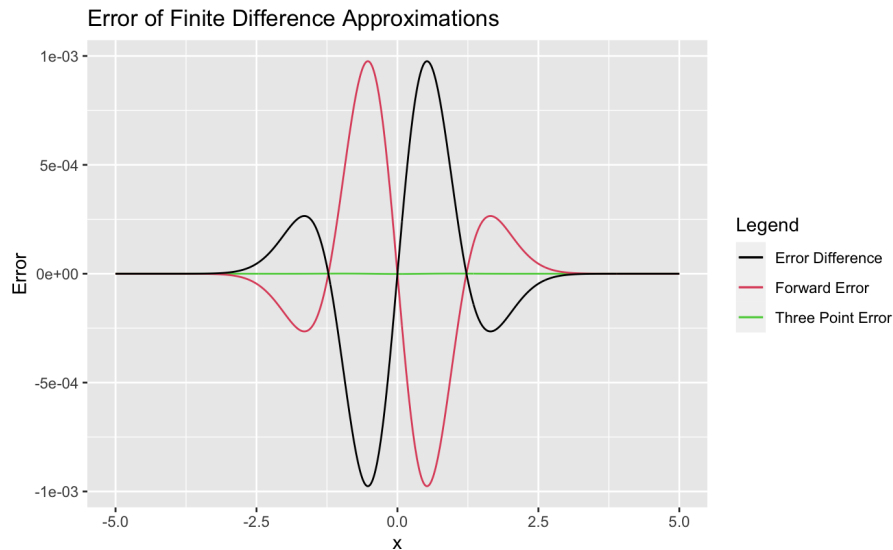


Figure 7: The error of the difference point methods as well as the difference between them. Three point centered difference error is magnitudes of order smaller.

4 Citation

Figure 3 is an image we used from <https://tutorial.math.lamar.edu/classes/calci/newtonsmethod.aspx>